

P08 – Allometric scaling in animals: is Kleiber’s law universal?

Field: Biology and Biophysics

Scientific question. Many biological quantities obey approximate power-law scaling relations. Does animal basal metabolic rate scale with body mass according to Kleiber’s law, $B \propto M^{3/4}$, or is the exponent different? Is a single exponent adequate for all animals, or do different taxonomic groups have different scaling exponents?

Data. Use the AnimalTraits database, a curated open database of terrestrial animal traits including body mass, metabolic rate and brain size:

- Dataset page: <https://animaltraits.org/>
- Zenodo record: <https://zenodo.org/records/6468938>
- Direct data file: <https://zenodo.org/records/6468938/files/observations.csv>
- Column documentation: <https://zenodo.org/records/6468938/files/column-documentation.csv>

Select N observations with both body mass and whole-body metabolic rate available. Retain only rows for which `metabolic rate - method` is `basal metabolic rate`. Use the standardized `metabolic rate` column, in watts, as the response variable. Analyze N animals together, or restrict the analysis to two taxonomic groups, birds and mammals.

Models. Let M_i be body mass and B_i metabolic rate. Work in log–log space:

$$x_i = \log M_i, \quad y_i = \log B_i.$$

- Model 1: Kleiber fixed-exponent model,

$$y_i = \alpha + \frac{3}{4}x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- Model 2: free allometric exponent,

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- Model 3: group-dependent scaling,

$$y_i = \alpha_{g[i]} + \beta_{g[i]}x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where $g[i]$ denotes the taxonomic group of observation i .

Required output.

1. Plot metabolic rate versus body mass on log–log axes, distinguishing the main taxonomic groups.
2. Infer the posterior distribution of the allometric exponent β .
3. Compute the posterior probability $P(\beta > 3/4 \mid \text{data})$ and report whether Kleiber’s exponent is inside the 95% credible interval.
4. Compare the fixed-exponent, free-exponent and group-dependent models.
5. Produce posterior predictive bands for metabolic rate over the observed body-mass range.
6. Predict the metabolic rate of an animal with mass $M = 1$ kg, including posterior uncertainty.

References.

- M. Kleiber, *The Fire of Life: An Introduction to Animal Energetics*, Wiley (1961).
- M. E. Herberstein et al., “AnimalTraits – a curated animal trait database for body mass, metabolic rate and brain size”, *Scientific Data* 9, 265 (2022).